

# Does an SMT Divergence Filter Add Edge to a Turtle-Soup Liquidity-Sweep Strategy?

A pre-registered, out-of-sample falsification across FX, metals and crypto

Candle Range Theory (CRT) · Turtle Soup · Smart-Money-Technique (SMT) divergence

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Independent quantitative research · methods and code released for full reproduction. All thresholds were fixed before results were seen; no parameter was tuned against the test set.

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## Summary

A popular discretionary playbook — a higher-timeframe CRT range, a Turtle-Soup liquidity sweep of a key level, and a cross-asset **SMT divergence** as confirmation, traded inside session killzones — was translated into a fully mechanical backtest and subjected to four **pre-registered** tests. The central question was deliberately narrow: **does the SMT divergence filter add genuine expectancy, or does it merely shrink the sample?**

Across all four tests the answer is the same: **no surviving edge**. The baseline Turtle-Soup pattern is net-negative after realistic costs everywhere. Wherever the SMT filter *did* show a statistically significant effect (gold vs the dollar, full-sample  $p \approx 0.013$ – $0.025$ ), the effect **failed to replicate out of sample**: it was concentrated entirely in the second half of 2025 and vanished on five untouched years of prior data (2020–2024,  $p = 0.46$ ) and on a different asset class (BTC/SOL,  $p = 0.78$ ). The apparent signal is the fingerprint of a single market regime, not a stable property of the strategy.

| # | Test                     | Data      | Full-sample p | OOS verdict                       |
|---|--------------------------|-----------|---------------|-----------------------------------|
| 1 | XAU/XAG (gold vs silver) | 2020–2024 | 0.46          | FALSIFIED – no effect             |
| 2 | XAU vs DXY-proxy         | 2025      | 0.013         | FALSIFIED – H1 fails ( $p=0.20$ ) |
| 3 | XAU vs synthetic DXY     | 2025      | 0.025         | FALSIFIED – H1 fails ( $p=0.13$ ) |
| 4 | BTC vs SOL (crypto)      | 2025–2026 | 0.78          | FALSIFIED – no effect             |

## Method

**The strategy, made mechanical.** Each execution bar (15 min) is mapped to the prior completed CRT range candle (4 h), which defines the key levels (CRT high / CRT low). A **Turtle Soup** short is registered when price pokes above the key level by a fixed buffer and then closes back below it within the same bar (mirror for longs). Entry is at that close; the stop sits beyond the sweep extreme; the target is a fixed 2R bracket. On FX/metals only London and NY-AM killzones are traded; crypto trades 24/7 (no sessions, no DST).

**The SMT filter.** A trade is flagged *SMT-confirmed* when a correlated instrument fails to confirm the sweep — e.g. gold prints a new high while the dollar fails to print a new low (divergence). Every test reports the baseline, the SMT-confirmed subset, and the non-SMT subset side by side, so the filter's marginal contribution is explicit rather than hidden.

**Anti-self-deception controls.** (1) Signals use only closed bars — no look-ahead. (2) Costs are charged per trade: 0.5 pip round-trip on FX, broker-style on metals, and a conservative **0.13% round-trip on crypto**. (3) A permutation test (10,000 shuffles of the SMT label) gives a distribution-free p-value for the SMT effect. (4) A bootstrap 95% CI tests whether net expectancy is robustly above zero. (5) Every threshold was locked *before* looking at results; no tuning against the test set.

**Pre-registered pass criterion (the decisive test).** For the SMT effect to count as real, the trade ledger is split chronologically at the calendar midpoint and the effect is recomputed independently in each half. The effect **survives only if both halves show a positive delta and  $p < 0.10$** . Anything else — a sign flip or either half non-significant — is recorded as falsified. A pass would mean the effect is *real and time-stable*, not necessarily profitable.

# Results

## 1 · The SMT filter does not produce a robust full-sample edge

On a permutation test, two of four configurations show a significant SMT effect (gold vs the dollar, via both a DXY proxy and a full synthetic DXY). Gold vs silver and crypto show nothing. Crucially, significance here means only that SMT-confirmed trades behave *differently* from the rest — not that they are *profitable*. Every SMT-only bootstrap CI either straddles or sits below zero.

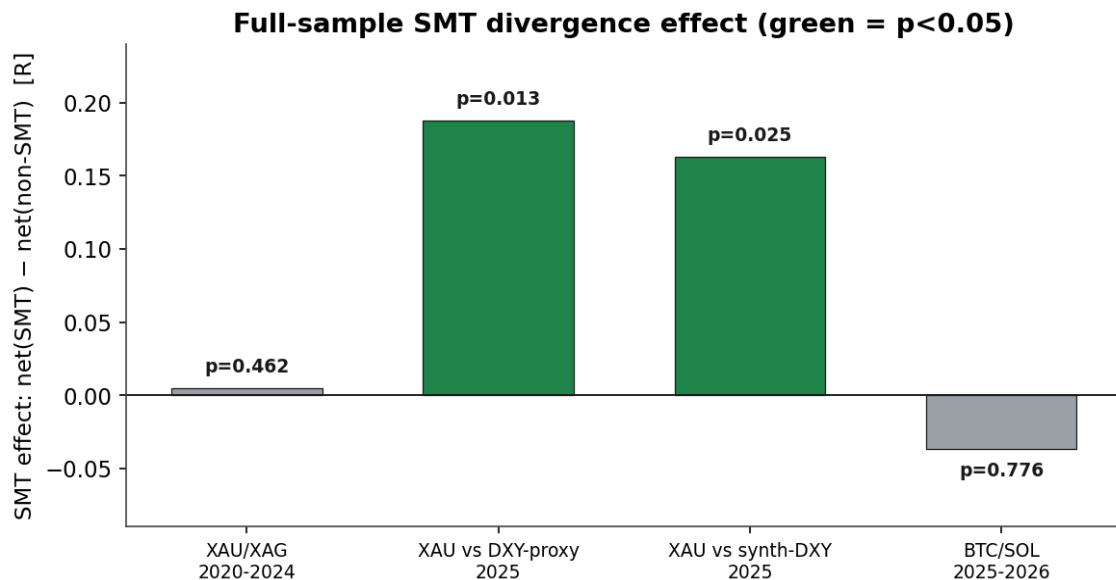


Figure 1. Full-sample SMT effect (R) with permutation p-values. Green bars are significant at  $p < 0.05$ ; both are gold-vs-dollar configurations.

## 2 · The effect does not replicate out of sample

This is the decisive figure. The two significant gold configurations owe their entire signal to the **second half of 2025**. In the first half the same filter is not significant ( $p = 0.20$  and  $0.13$ ), failing the pre-registered criterion. On the five independent years of 2020–2024 the effect is essentially zero ( $p = 0.46$ ), and on crypto it is negative in both halves. A genuine cross-market edge would appear in data the discovery period never touched; it does not.

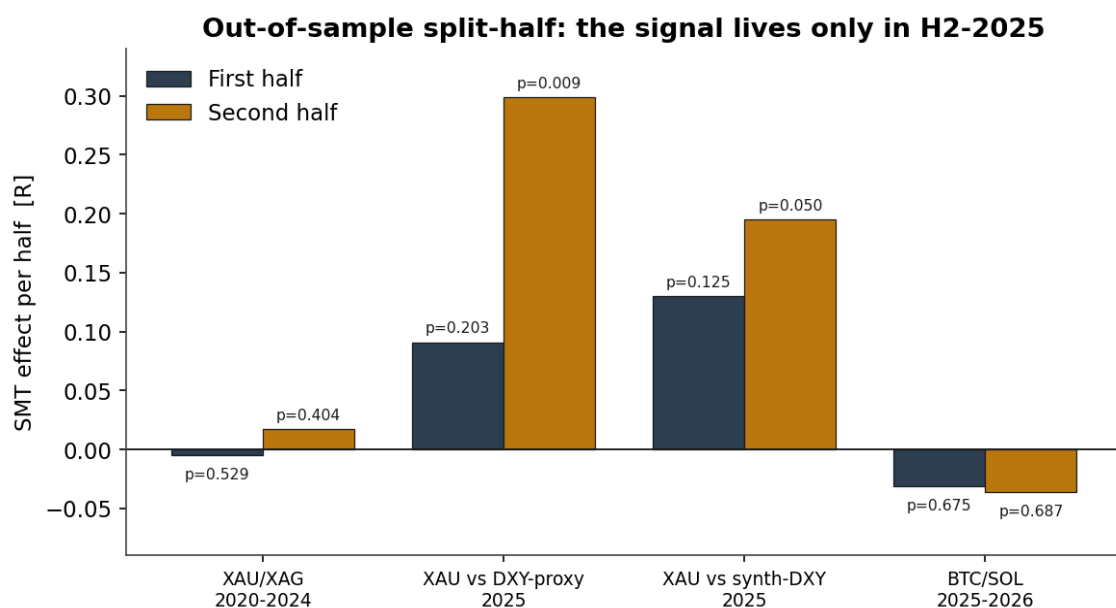


Figure 2. Out-of-sample split-half. The signal is localized to H2-2025 and disappears everywhere else — the signature of a single regime, not a stable edge.

### 3 · Nothing is profitable after costs

Across all four tests and every cell — baseline, SMT, non-SMT — net expectancy is negative. Even the single most favourable cell (gold vs dollar, second half of 2025) only reaches roughly break-even before costs and turns negative after them.

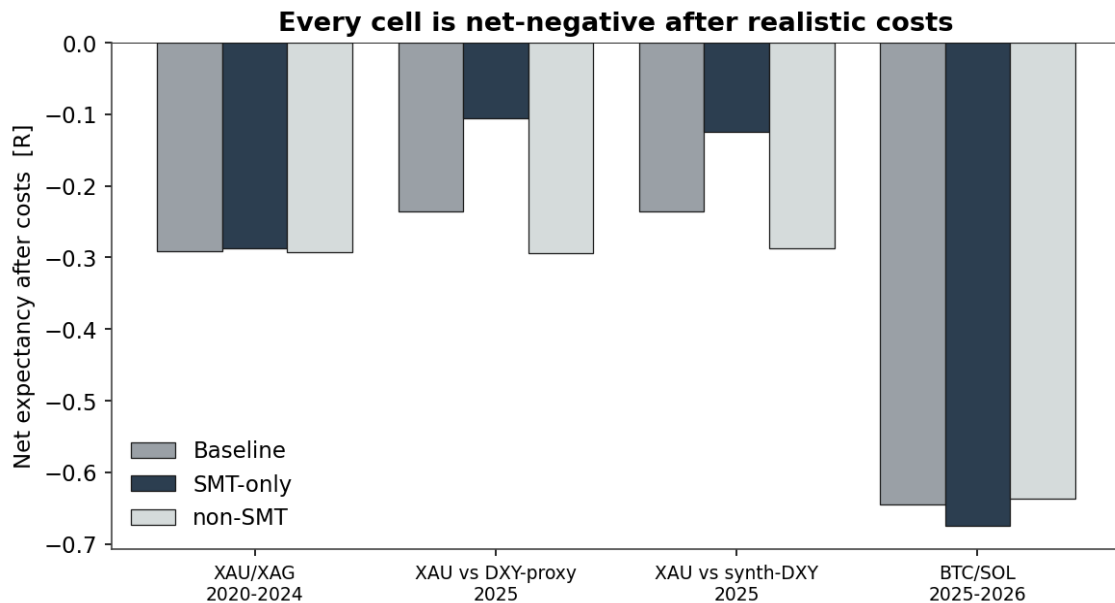


Figure 3. Net expectancy after costs, per cell. Not one configuration clears zero.

### 4 · Costs do most of the killing

The gap between gross and net expectancy is large and, on crypto, decisive: a gross of  $-0.07R$  becomes  $-0.65R$  after a 0.13% round-trip — costs alone subtract  $\sim 0.57R$  per trade. On a 15-minute sweep pattern the per-trade move is simply smaller than the friction. This is the same lesson that ended the earlier crypto order-flow work, now confirmed on a different pattern family.

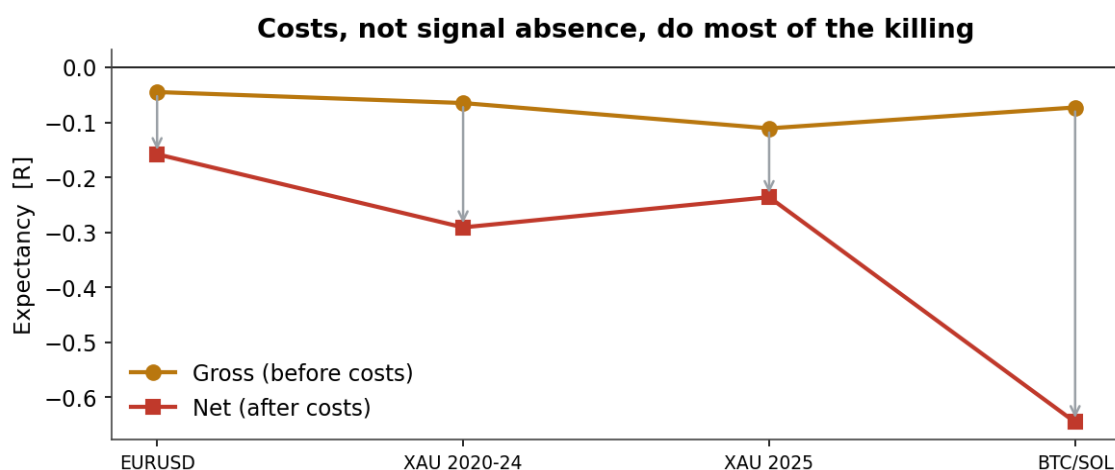


Figure 4. Gross vs net expectancy. The arrow length is the per-trade cost drag.

## Interpretation

The honest reading is not “the indicators are noise.” The SMT divergence between gold and the dollar carries real information in a specific window — the permutation p-values are not flukes within that window. The problem is **stability**: the information does not generalize backward in time or across assets. A strategy whose edge exists only in the period where it was discovered is, for trading purposes,

indistinguishable from one with no edge — and is actively dangerous, because it backtests beautifully and then fails live.

This is also a clean illustration of why a single in-sample  $p = 0.02$  is not evidence. Three of these tests produced a “significant” full-sample result at some stage; all three dissolved under an out-of-sample split that was specified in advance. The pre-registered criterion did exactly its job: it stopped a tempting +0.07R second-half cell from being mistaken for a discovery.

## Limitations

Costs are modelled, not broker-exact; the metals cost assumptions in particular are conservative placeholders. Killzone hours on FX/metals are not yet DST-adjusted, which can blur a session-specific edge — though no session showed positive net expectancy regardless. Exits use a fixed 2R bracket; a target tied to the opposing CRT liquidity could change the distribution, but not the central finding that the SMT *filter* adds no stable edge. The synthetic DXY is an OHLC reconstruction from six pairs; a DXY proxy and the full index agreed closely, which is reassuring.

## Reproducibility

All code is released: the backtest engine (CRT levels, Turtle-Soup detection, SMT modes, fixed-R simulation, permutation + bootstrap + pre-registered split-half OOS), the HistData converter with year filtering, the DXY-proxy builder, and the full synthetic-DXY builder. Data are free 1-minute bars from HistData.com (FX/metals) and Binance OHLC (crypto). Every result in this report regenerates from a single command per test.

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Methodology over outcome. This study reports a negative result because the data produced one. The value is in the process: fixed thresholds, costs taken seriously, an out-of-sample test specified before the numbers were seen, and a willingness to reject a tempting cell. That is the discipline a trading system is supposed to survive.